

TEBM28C10-4B Datasheet

1. Overview

The TEBM28C10-4B Balanced Mode Radiator (BMR®) is an audio drive unit with an extended frequency range and wide directivity. The small form-factor is ideally suited for compact products that require a full-range drive unit, room filling sound and a high-performance acoustic solution.

- Power Handling: 10 W
- Nominal Impedance: 3.8 Ω
- 55 mm Dia x 24 mm Depth
- 58 g Mass

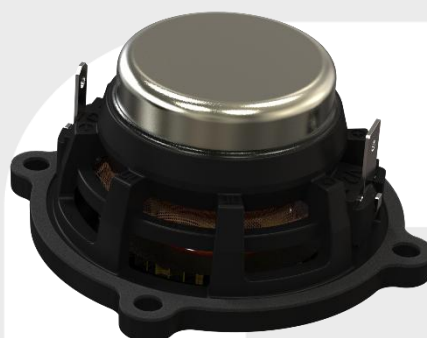


Figure 1.1

**Product code and manufacture date is printed at the back of the return cup*

2. Applications

- Full-range Compact Systems
- Conferencing Systems
- Bluetooth Audio
- Smart Speakers
- Gaming Devices
- IoT Devices

3. Specifications

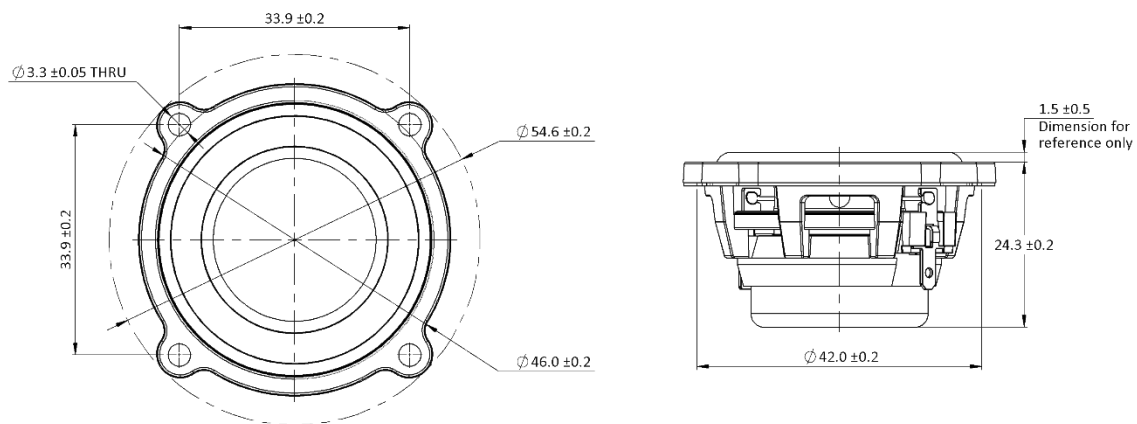
Transducer Characteristics			Parameter	Nominal	Unit
Frequency Response (± 6 dB)	150 Hz ~ 16.5 kHz		Fs	145	Hz
Speaker Sensitivity (1 Watt / 1 meter)	80	dB	Sd	8.55	cm ²
Rated Maximum SPL (1 Meter)	90	dB	Mms	1.19	g
Speaker Nominal Impedance	3.8	Ω	Cms	1.0	mm/N
Voice Coil Diameter	19.05	mm	Rms	0.31	N*s/m
Voice Coil Material	Copper		Re	3.8	Ω
Diaphragm Material	Doped Paper Composite		Bl	2.9	Tm
			Le	0.10	mH
			Qts	0.44	

3.1. Operating Conditions

Rated Noise Power (100 hours) <i>IEC268 Pink noise with 2nd order high-pass filter at 150Hz, 6dB crest factor, transducer in free air, ambient conditions – normal temperature and pressure</i>	: 10 W
Operating Ambient Temperature Range	: –20 to +55 °C
Max Linear Excursion*	: 2.8 mm Peak to peak
Mechanical Excursion Limit	: 8 mm Peak to peak
Max Surround Frontal Movement	: 2.2 mm

**From Klippel LSI*

3.2. Product Dimension



Note:

- Volume Displacement: 13 cc
- All dimensions are in mm

Figure 3.2.1 – External product dimensions

3.3. On-Axis SPL and Impedance (Measured)

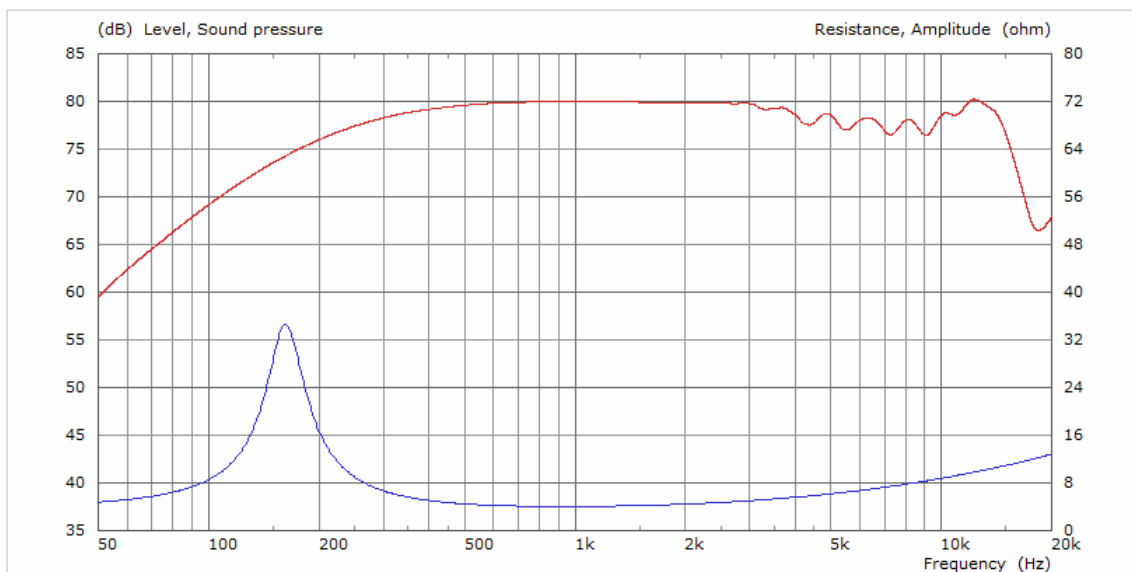


Figure 3.3.1 – Red: On-Axis SPL at 1W/1m (1/3-octave smoothed/spliced*/anechoic). Blue: Electrical Impedance

3.4. Sound Power Response (Measured over 0 – 90°)

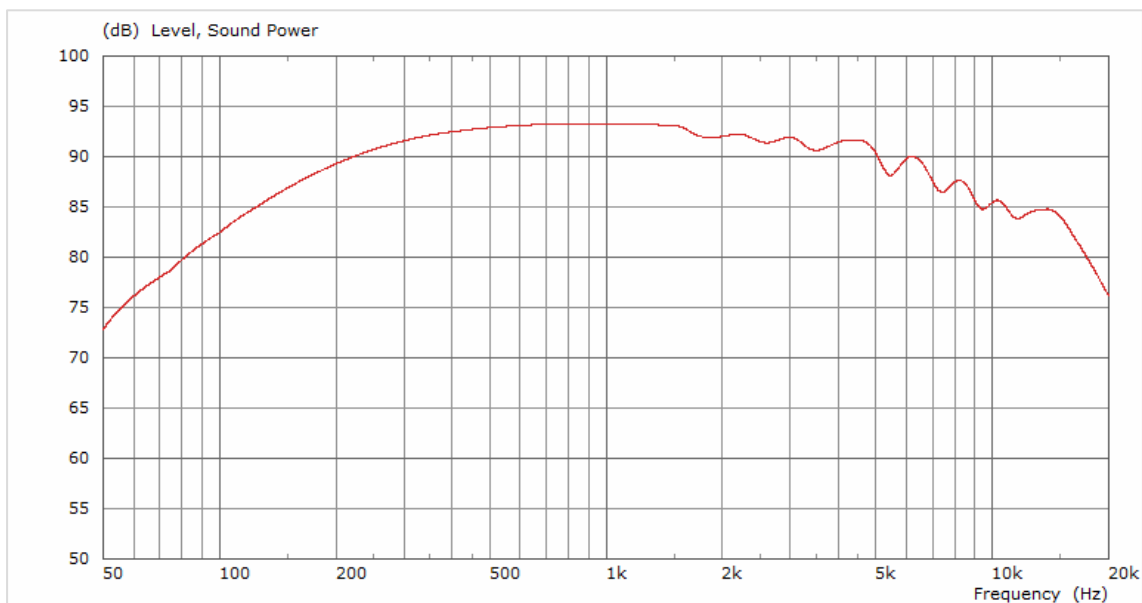


Figure 3.4.1 – Sound power calculated from SPL measurements, 1W/1m (1/3-octave smoothed/spliced*)

**Anechoic acoustic measurement spliced to low frequency response derived from diaphragm scan using Polytec PSV500 scanning vibrometer*

3.5. Polar Response (Measured)

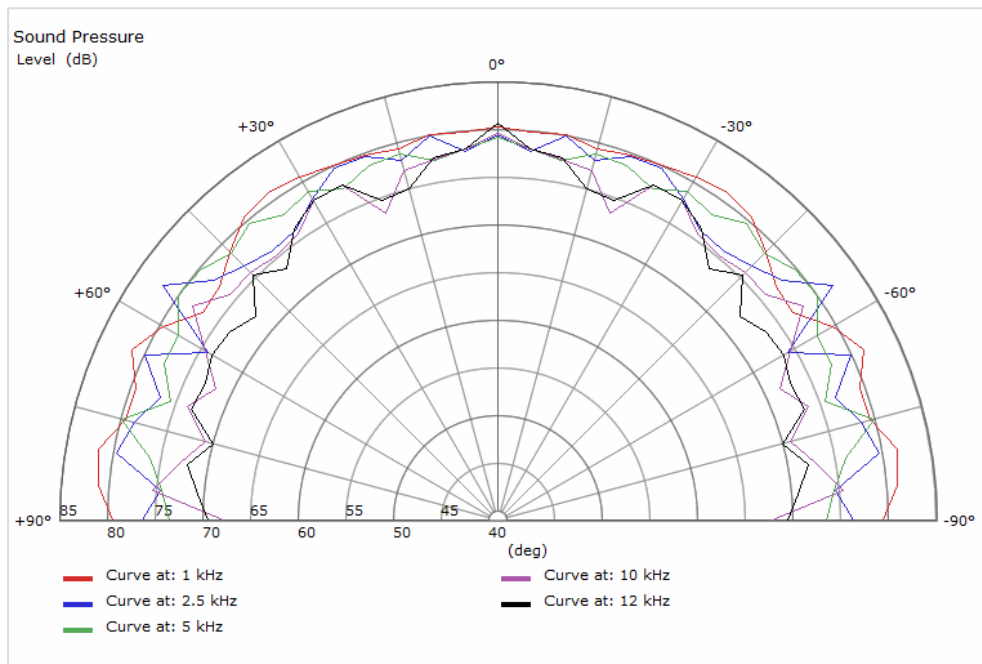


Figure 3.5.1 – Polar response, angle/ dB SPL, 1W/1m (1/3-octave smoothed / anechoic)

4. Appendix

4.1. Klippel LSI

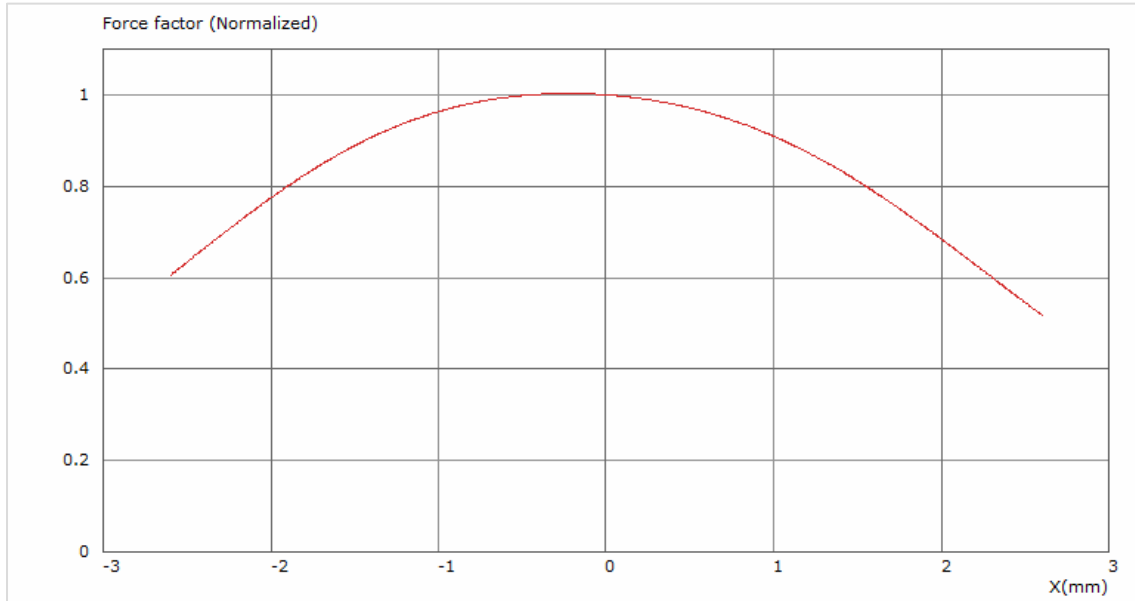


Figure 4.1.1 – Normalized BL (x)

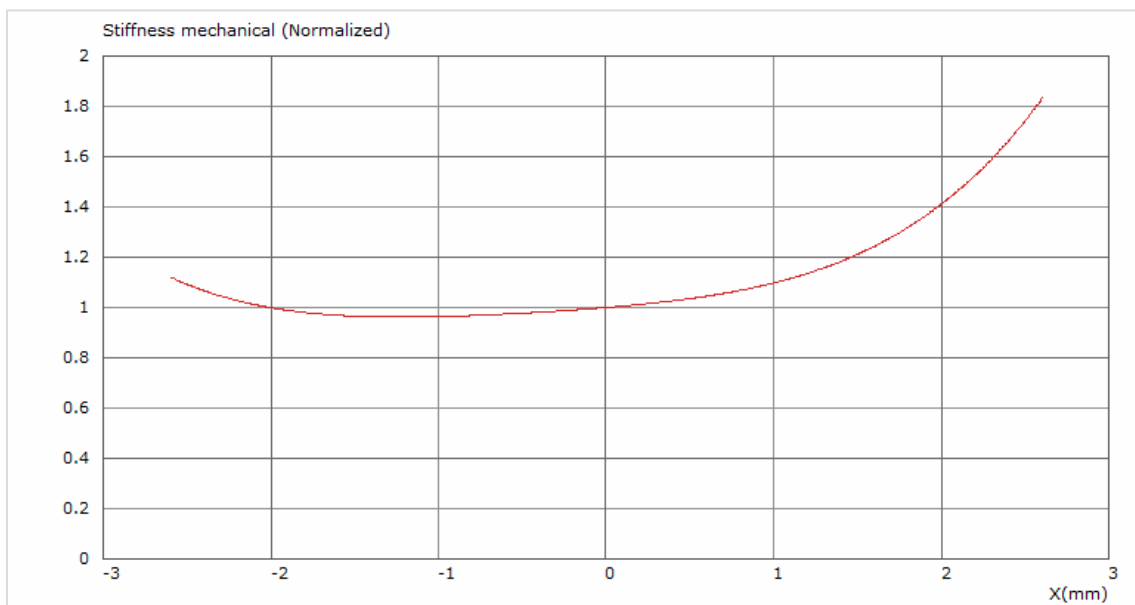


Figure 4.2.1 – Normalized Kms (x)